OO-M-0038K(Army-GL)
October 25, 1985
INTERIM REVISION OF
OO-M-38J
November 23, 1977

INTERIM FEDERAL SPECIFICATION

MIXING MACHINES, FOOD, ELECTRIC

(VERTICAL, COMMERCIAL TYPE, BAKERY AND KITCHEN)

This interim Federal specification was developed by the U.S. Army Natick Research and Development Center (GL), Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim Federal specification as a valid exception to OO-M-38J, dated November 23, 1977.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial, electric motor-driven, vertical food mixers. The machines shall be adaptable for mixing, beating, and whipping operations, slicing vegetables, and meat chipping (grinding) (see 6.1).

1.2 Classification. The food mixing machines shall be of the following models, sizes, and styles as specified (see 6.2).

MODEL I, (bench mounted)

	Bowl capacity quarts		Motor horsepower
Size	Minimum	Maximum	Minimum
12	12	15	1/4
20	20	25	1/2
	12	15	-

FSC 7320

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MODEL II, (floor mounted)

Size	Bowl capacity quarts Minimum	Maximum	Motor horsepower Minimum
30	30 20	36 25	3/4 -
60	60 30	70 36	1-1/2 -
80	80 40	90 --	2
140	140 60	160 80	5 -

Style 1 - Standard height
Style 2 - Tall column (size 80, only)

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
QQ-A-601	-	Aluminum Alloy Sand Castings
TT-P-664	-	Primer, Coating, Synthetic Rust-Inhibiting, Lacquer-Resisting
MMM-A-260	-	Adhesive, Water-Resistant (For Sealing Waterproofed Paper)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-60	-	Tape: Packaging. Waterproof

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Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	- Preservation Methods Of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic and Associated Equipment)
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc., (ANSI)

S1.13 - Methods for the Measurement of Sound Pressure Levels

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate Sheet and Strip
A 412 - Stainless and Heat-Resisting Chromium-Nickel-Manganese Steel Plate, Sheet, and Strip
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 10103.)

National Sanitation Foundation (NSF)

No. 2 - Food Service Equipment
Listing of Food Service Equipment
No. 8 - Commercial Powered Food Preparation Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, 3475 Plymouth Road, Ann Arbor, MI 48105.)

Rubber Manufacturer's Association (RMA) Publications

Engineering Standards - Drives Using Classical Multiple V-Belts (A, B, C, D and E Cross Sections)
- Drives Using Synchronous Belts (XL, L, H, XH, and XXH Belt Sections)

(Application for copies should be addressed to the Rubber Manufacturer's Association, 444 Madison Ave., New York, NY 10022.)

Underwriters Laboratories Inc.

UL 73 - Motor-Operated Appliances

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The mixing machines shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturers standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The mixing machines shall comply with the applicable requirements of National Sanitation Foundation (NSF) standards numbers 2 and 8 and Underwriters Laboratories Inc. (UL) standard number 73. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.3).

3.3 Material and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial food mixing machines provided the complete item complies with all provisions of this specification (see 6.4).

3.3.1 Castings. Aluminum alloy castings for bases and pedestals shall be alloy 355.0 or 319.0 conforming to QQ-A-601. Aluminum alloy castings for agitators shall be 713.0, 535.0, D712.0, or alloy 355.0 conforming to QQ-A-601. Castings shall be of uniform quality, shall not have blowholes, be porous, have hard spots nor be cracked. Aluminum castings shall be polished to a smooth sanitary finish.

3.3.2 Stainless steel. Stainless steel for mixing bowls shall conform to type 201 of ASTM A 412 or types 301, 302 or 304 of ASTM A 167.

3.3.3 Bearings. Bearings shall be replaceable and of a size and design to offset any radial and end-thrust loads developed by the moving parts.

3.4 Design. Food mixing machines shall be model, size, style, and bowl capacities as stated in 1.2 and shall be designed for mixing and blending food ingredients by means of interchangeable agitators and whips. The machine shall be driven by an electric motor. Beater speed shall be adjustable. All

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parts subject to wear shall be accessible for adjustment and replacement. Machine design shall allow removal of mixing bowls with contents. When tested as specified in 4.4.2.2, the food mixing machines shall be capable of thoroughly mixing fully developed batches of bread dough from the quantities of ingredients specified in 4.4.2.1, without exceeding a motor temperature rise of 45°C for the insulated motor and without damage to components.

3.4.1 Interchangeability. All replaceable parts shall be constructed to definite commercial standards, tolerances, and clearances in order that any part of a particular model, size, or style machine may be replaced.

3.5 Illustrations. The illustrations identified herein as figures 1 and 2 are for identification purposes only and are not intended to preclude the purchase of food mixing machines otherwise conforming to this specification.

3.6 Construction.

3.6.1 Drive. The machine shall be driven by an electric motor connected through gearing, geared timing belt, V-belt(s), or enclosed chain drive. Gearing, when used, shall be fully enclosed, shall have machine cut teeth, and shall be quiet in operation. Cast unmachined gears are not acceptable. Guards shall be provided to protect the operator against contact with projections or moving parts which are exposed outside the main unit housing. Plastic material shall not be used for any guards or housing. All gearing shall be accessible and replaceable. Model II machines shall be equipped with a clutch, shock absorber, V-belt, or geared timing belt drive to minimize the shock of starting the beater and shifting gears. The clutch for model II machine shall positively disengage the drive spindle from the driving unit when the speed selector is in the neutral position. Where V-belt is used on model II machines, the drive shall be either a multiple V-belt drive that shall be in accordance with RMA Engineering Standards for Drives Using Classical Multiple V-Belts, with a service factor of at least 1.5 or a variable speed transmission or drive consisting of adjustable pulleys with endless cord V-belts. Where geared timing belt is used for model II machines, the belt shall be in accordance with RMA Engineering Standards for Drives Using Synchronous Belts. There shall be no evidence of deformation or wear of the complete drive mechanism when tested as specified in 4.4.2.

3.6.2 Speed selector. A speed selector with appropriate markings for each speed setting shall be provided to change beater speeds. The speed selector shall not move from the selected setting while the machine is in operation. Gear-driven machines shall be provided with a positive engagement - disengagement mechanism for all speed settings including neutral position if one is provided. The speed selector shall be capable of changing speeds by one hand manipulation (see 4.4.1 and 4.4.2).

3.6.2.1 Speed selector interlock. When one or more neutral positions are furnished for the speed selector, a speed selector interlock shall also be provided. The interlock shall automatically prevent movement of the speed

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selector when the speed selector is placed in any neutral position(s). Release of the speed selector interlock shall require an action by the operator different from the action required to move the speed selector. There shall be no evidence of deformation or wear of the interlock mechanism when tested as specified in 4.4.1 and 4.4.2.2.

3.6.3 Drive spindle. The drive spindle shall be vertically mounted and shall be securely held in place. Agitators shall fit securely and shall be capable of being removed and replaced. The drive spindle shall be equipped with planetary gearing to effect two separate rotating movements to the agitators, in opposite directions, simultaneously. The drive spindle may be equipped with gearing to allow the agitator to rotate in the same direction as the planetary head.

3.6.4 Electrical system.

3.6.4.1 Motors. Alternating current motors shall conform to CC-M-1807 and with any additional requirements specified in this specification. Motors shall have the minimum horsepower ratings indicated in 1.2. Unless otherwise specified (see 6.2), model I food mixing machines shall be designed for operations on a 120-volt, 60 Hertz (Hz), single phase system. Unless otherwise specified (see 6.2), model II food mixing machines shall be designed for operation on a 208-volt, 60 Hz, three-phase, system. When 50 Hz is specified, all motors shall be designated to operate on 50 Hz. Single phase motors shall be capacitor start, induction run, with only one contact. Three-phase motors shall be squirrel-cage, normal starting torque, normal starting current type. Machines with three-phase motors shall be provided with a corrosion-resistant metal plate, label or decalcomania (decal) showing the direction of rotation of the beater spindle. Unless otherwise specified (see 6.2), motors, when fully installed in the machine, shall have the following characteristics:

TABLE I. Motor design characteristics and requirements

Design characteristics	Requirement
Enclosure	Splashproof, totally enclosed, or dripproof
Speed	Constant
Duty	Continuous or 1-hour varying
Ambient temperature	40°C
Temperature rise	As specified in 3.4

3.6.4.2 Windings. When direct current is specified, model I machines shall be compound wound. Direct current motors for model II machines shall be shunt or compound wound.

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3.6.4.3 Cord and attachment plug. Unless otherwise specified (see 6.2), model I machines shall be furnished with a flexible, oil-resistant, rubber covered, type SO or SJO cord with a UL approved, 2 pole, 3 wire grounding type plug with dead front construction. The cord shall be a 3-wire grounding type and shall extend not less than 6 feet beyond the machine. Model II machines shall have provisions for permanent electrical connection.

3.6.5 Electromagnetic compatibility. When specified (see 6.2), mixing machines and table procured under this specification shall be designed and equipped to meet the electromagnetic interference control requirements and test limits for part 10, class C3, group I equipment as specified in MIL-STD-461, when tested as specified in 4.4.5.

3.6.6 Fungus and moisture resistance of electrical components. When specified (see 6.2), mixing machines shall be treated to resist fungus and moisture as follows:

(a) Circuit elements which have a temperature rise of not more than 75°F when operating at full load, shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, terminal blocks, capacitors, and coils.

(b) Electrical components not requiring fungus moisture resistance: Electrical components such as switches, fuses, and contacts, shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated.

3.6.7 Starters. Motors for model I machines shall be provided with fully enclosed, quick break, manual starters operated by indicating snap switch. Model II machines shall be provided with starters with automatic-reset or manual reset thermal overload and low voltage protection. On machines provided with a timer, a magnetic, thermal overload, automatic reset type switch is acceptable. The enclosure shall be drip-proof. The rating shall be based on an ambient of 50°C with a maximum temperature rise of 55°C as measured by the thermometer method.

3.6.8 Noise limits. The food mixing machines shall not exceed 85 decibels A-weighted at the normal operator position when tested as specified in 4.4.6.

3.6.9 Mixing bowls. The mixing bowls provided with each food mixing machine shall be within the capacity range listed under 1.2, when tested as specified in 4.4.7. Two mixing bowls shall be provided with each size mixing machine as specified in 1.2, except for size 12 mixing machine. The size 12 mixing machine shall be provided with one mixing bowl. When additional bowls are required, the size and quantity shall be specified (see 6.2). Bowls shall be formed from a single piece of the material specified in 3.3.2. Starting material thickness shall be nominal 0.0625 inch for bowls of 60 quart capacity and less and shall be nominal 0.0781 inch for bowls over 60 quart capacity. Minimum thickness at any point of the bowls after forming shall be not less

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than 0.050 inch for bowls of 60 quart capacity and less, and not less than 0.062 inch for bowls over 60 quart capacity. The contractor shall furnish a certificate stating the minimum thicknesses meet the specification requirements. Each bowl shall be provided with two handles fastened to the bowl body. Bowl finish shall be as specified in 3.12.2. One splash cover for each 80 quart and 140 quart size bowl shall be provided for use during operation and shall be fabricated of same material as the bowl except when specified (see 6.2), the splash cover for each 80 quart and 140 quart size bowl shall be molded of a high impact grade of acrylonitrile butadiene styrene material which is inherently resistant to cooking oils and heat encountered in normal operation. The design of the plastic cover shall be the manufacturers commercial design and shall be NSF approved. Bowls shall have three lugs to match the lug positions on the bowl support (see 3.6.11).

3.6.10 Agitators. A batter beater, sweet dough beater, dough hook, pastry knife, wing whip, and wire whip suitable for use with each bowl size specified shall be provided with each food mixing machine. Beaters and hooks shall be fabricated of aluminum alloy specified in 3.3.1, stainless steel specified in 3.3.2, or a combination of both. Wire and wing whips shall be made entirely of stainless steel wires. Pastry knives shall be made entirely of aluminum alloy or stainless steel, or shall have an aluminum alloy head with a stainless steel blade. Agitators shall be designed to permit cleaning without disassembly.

3.6.11 Mixing bowl support. Each food mixing machine shall be provided with an operating mechanism for raising and lowering the mixing bowl support. Mixing bowl support mechanisms shall be capable of raising and lowering a load of dough when tested as specified in 4.4.4. The adjusting mechanism shall be self locking, or shall be provided with a positive lock so the moving members can be manually locked in position. Size 60 and larger machines shall have provisions for locking the bowl at any position from operating to lowest position. Model I and model II, sizes 30, 60, and 80 shall have a hand operated raising and lowering device. Size 30 and smaller machines shall have locking features as stated, or shall have provisions for locking the bowl in operating position only. An automatic stop shall be provided to prevent the agitator from contacting the bowl, when proper bowls and agitators are used together. The adjusting mechanism for model II machines shall be capable of lifting the mixing bowl from a dolly and raising it to operating position on the machine. Bowl support shall have three lug positions matching the lugs on the bowl (3.6.9).

3.6.12 Lubrication. All wearing parts of the food mixers shall be provided with means for lubrication or permanently lubricated and sealed. Oil seals shall contain the lubricant within the fixed limits. Gear housings shall have provisions for lubrication and shall allow lubricant to circulate around gears.

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3.6.13 Lubricating instructions. The food mixing machines shall be provided with a corrosion-resisting metal lubrication chart affixed to the item indicating all points of the equipment requiring lubrication, the type and grade of lubricant required, and the time interval when lubrication should be accomplished.

3.6.14 Fastening devices. All screws, pins, and bolts shall be installed with means for preventing loss of tightness and adjustment. All such components, when subject to removal or replacement, shall not be swaged, peened, staked, nor otherwise permanently deformed to prevent removal.

3.6.15 Identification marking. Identification, operation instructions and speed selector settings shall be permanently and legibly marked directly on the item or on an aluminum, brass or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or photographically marked. The plate thickness shall be not less than 0.003 inch.

3.6.15.1 Identification plate. Each mixing machine identification plate shall include the manufacturer's model and serial number, name, tradename or trademark, of such known character as to be readily identifiable. In addition, such information as required by UL 73 for the mixing machine shall be included on the plate.

3.6.15.2 Users instruction plate. The food mixing machines shall be provided with an instruction plate in a location visible to the operator. The words "MACHINE, FOOD MIXING" shall be not less than 3/16 inch high and the words "USER'S INSTRUCTIONS", shall be not less than 1/8 inch high. All other characters shall be not less than 3/32 inch high. The plate shall be of sufficient overall size to contain the following information: The plate shall have the following information etched or die-stamped thereon with legible letters.

MACHINE, FOOD MIXING

USER'S INSTRUCTIONS

1. Lock bowl in position.
2. Use proper agitator for ingredients being worked.
3. Use clutch when shifting speeds. If no clutch, stop motor between speed changes. 1/
4. Fill bowl at lowest position. Raise bowl slowly as food breaks up.
5. Do not use high speeds for heavy loads. Start mixing yeast raised dough in low speed. Finish dough in second speed.
   CAUTION:
   Do not use higher speeds for these doughs 1/
6. Use attachments at the manufacturer's recommended speed.
7. Do not put hands, spoons, nor utensils on bowl while machine is in operation.

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8. Wash bowl and agitator immediately after using. If egg mixture or flour batters were used, apply cold water before washing with hot water.
9. Dry agitators thoroughly and hang up to prevent damage.
10. Dry bowl thoroughly.
11. Do not drag bowl over floor.
12. Wash body of machine with warm water and soap. Do not rinse with hose.

1/ Omit item 3 and third and fourth sentence of item 5 on variable speed drive machines.

3.7 Detailed construction. Speeds specified in 3.7.1.4 through 3.7.2.5 shall be defined as the spindle speed relative to the mixing bowl.

3.7.1 Model I (bench mounted). Each model I food mixing machine shall be provided with mixing bowls and agitators as specified in 3.6.9 and 3.6.10. Intermediate spindle speeds shall be approximately evenly divided between the low and the high speeds. All spindle speeds shall be tested as specified in 4.4.1.

3.7.1.1. Vertical column. The vertical column shall be cast iron, cast or welded steel, stainless steel, or the cast aluminum alloy specified in 3.3.1. The vertical column, base, spindle support, and the transmission housing may be formed as an integral unit or they may be formed as separate units and joined by welding, bolting, or both.

3.7.1.2 Base. The base shall be constructed of material as specified for vertical column in 3.7.1.1 and shall be constructed to provide stability to the machine. Means shall be provided for bolting the model I machines to a bench so that when operated at full speed, as specified in 4.4.1, there will be no noticeable vibration of the machine. Unless otherwise specified (see 6.2), the method of bolting the machines shall be at the option of the manufacturer. Navy ship board installation mounting shall be as specified in 3.11.

3.7.1.3 Size 12. The size 12 food mixing machine shall be suitable for operation with a bowl within a 12 to 15 quart capacity range, and shall have not less than three speeds divided as specified in 3.7.1. Unless otherwise specified (see 6.2), the spindle rpm in low speed shall be not greater than 120 and the rpm in high speed shall be not less than 300.

3.7.1.4 Size 20. The size 20 food mixing machine shall be suitable for operation with a bowl within a 20 to 25 quart capacity range, and shall have not less than three speeds divided as specified in 3.7.1. The size 20 mixer shall be capable of mixing 25 pounds of bread dough based on 14 pounds of flour when tested as specified in 4.4.2. Unless otherwise specified (see 6.2), the spindle rpm in low speed shall be not greater than 120 and the rpm in high speed shall be not less than 300.

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3.7.1.5 Table. When specified (see 6.2), a table shall be provided for the model I, size 20 food mixing machine. The table shall be of open base, welded construction with nominal overall dimensions of 30 inches deep, 24 inches wide and 24 inches high with a shelf located a minimum of 9-1/2 inches above the floor. The frame shall be constructed of not less than nominal 1-1/4 inch type 302 or 304 stainless steel tube having a minimum wall thickness of 0.083 inches. The tubular upright forming the legs shall be welded at the top and below to the horizontal tubular cross braces forming a one piece welded unit. The horizontal cross braces shall support the shelf. The table shall be provided with a stainless steel sheet top and shelf not thinner than 0.065 inches thick. Reinforcing shall be provided under the top platform to rigidly support the food mixing machine. Edges of the top and shelf shall be flanged or rolled down over the top and lower horizontal cross braces of the frame and shall be secured to allow ease of cleaning. Each leg shall be provided with an adjustable leveling device. When specified (see 6.2), casters shall be provided with each table. Caster shall be of the ball bearing swivel type. Two casters shall be furnished with foot operated brakes. The combined height of table and mixing machine shall not exceed 55 inches. The table shall conform to the applicable requirements of NSF Standard No. 2. The table shall be capable of supporting a load of 250 pounds without deformation, when tested as specified in 4.4.9.

3.7.2 Model II (floor mounted). Each model II food mixing machine shall be provided with mixing bowls and agitators as specified in 3.6.10. Sizes 60, 80, and 140 food mixing machines shall be provided with a bowl dolly, and bowl dolly adapter conforming to the requirements of 3.10.2 and 3.10.6, respectively. Intermediate spindle speeds shall be approximately evenly divided between the high and low speeds. All spindle speeds shall be tested as specified in 4.4.1.

3.7.2.1 Vertical column and base. The vertical column and base shall be constructed as specified in 3.7.1.1 and 3.7.1.2 respectively, except means shall be provided for bolting the machine to the floor. The vertical column of the style 2 food mixing machine shall be of a height that will allow insertion and removal of an 80 quart bowl without removal of beaters or agitators. For Navy ship board installation mounting, see 3.11.

3.7.2.2 Size 30. The size 30 food mixing machines shall be suitable for operation with a bowl within a 30 to 36 quart capacity range and shall have not less than three speeds. The mixer shall be capable of mixing 40 pounds of bread dough based on 22 pounds of flour, when tested as specified in 4.4.2. Unless otherwise specified (see 6.2), the spindle rpm in low speed shall be not greater than 100 rpm in high speed shall be not less than 300.

3.7.2.3 Size 60. The size 60 food mixing machines shall be suitable for operation with a bowl within a 60 to 70 quart capacity range and shall have not less than four speeds. The mixer shall be capable of mixing 75 pounds of bread dough based on 42 pounds of flour when tested as specified in 4.4.2. Unless otherwise specified (see 6.2), the spindle rpm in low speed shall be not greater than 80 and the rpm in high speed shall be not less than 280.

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3.7.2.4 Size 80. The size 80 food mixing machines shall be suitable for operation with a bowl within an 80 to 90 quart capacity range. The machine shall have not less than four speeds. Unless otherwise specified (see 6.2), the spindle rpm in low speed shall be not greater than 80 and the rpm in high speed shall be not less than 300. The mixer shall be capable of mixing 100 pounds of bread dough, based on 55 pounds of flour, when tested as specified in 4.4.2.

3.7.2.5 Size 140. The size 140 food mixing machines shall be suitable for operation with a bowl within a 140 to 160 quart capacity range, and shall have not less than four speeds. The mixer shall be capable of mixing 180 pounds of bread dough, based on 100 pounds of flour, when tested as specified in 4.4.2. Unless otherwise specified (see 6.2), the rpm in low speed shall be not greater than 55 and the spindle rpm in high speed shall be not less than 260. Adjustment of the mixing bowl support or drive shall be accomplished by an electric motor.

3.8 Timer. The model II machines shall be provided with 15 minute (minimum) timer. The timer shall automatically stop the machine in any speed after reaching a preset time. The timer design shall include a nontiming position to permit continuous operation with vegetable slicer or meat chopper.

3.9 Operating hub. Unless otherwise specified (see 6.2), food mixing machines shall be provided with a hub drive for operating a vegetable slicing attachment or a meat chopping attachment (see 3.10.4 and 3.10.5, respectively).

3.10 Accessories and attachments. As required hereinafter, the food mixing machines shall be provided with the following additional accessories and attachments.

3.10.1 Mixing bowl extension. A mixing bowl extension shall be provided for the 60 quart and 80 quart capacity mixing bowl for size 60 and 80 mixing machines, respectively. The mixing bowl extension shall fit the bowl snugly and shall be of sufficient height to prevent "throw out" while mixing. The mixing bowl extension shall be of the same material as the bowl. Finish shall be as specified in 3.12.1.

3.10.2 Dollies. Unless otherwise specified (see 6.2), a dolly for each size bowl specified shall be provided with model II, sizes 60, 80, and 140 machines. The dollies shall be suitable for receiving, holding, and rolling mixing bowls to and from their position at the machine without manually lifting the bowl with its contents. Dollies shall be fabricated of aluminum alloy and finished as specified in 3.12.4. When tested as specified in 4.4.8, the dolly shall show no evidence of damage or permanent deformation. Dollies for carrying mixing bowls of 60 quart capacity shall have at least three ball bearing swivel casters. Dollies for carrying mixing bowls of 80 quart capacity and over shall have four ball bearing swivel casters. The casters shall be stud mounted or truck type, rubber tired with metal wheels or molded composition wheels with a nominal 3/4 or 7/8 inch tread width. At least two casters on all dollies shall be provided with locking features.

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3.10.3 Colanders. A colander set shall be provided for the size 20 and size 30 mixing machines. The colander set shall consist of a sieve holder, three sieves (fine, medium, and coarse), a roller beater or scraper beater, a drain bowl or mixing bowl and necessary parts for assembling together and to the mixer. The sieve holder shall fit the drain bowl furnished as a component of the colander set or the size mixing bowl furnished with the mixer. When specified (see 6.2), an additional bowl shall be furnished and shall be suitable for use with the size colander provided. The colander shall be adaptable for straining soups, making purees and sauces, and other similar work. The sieve holder and bowl shall be stainless steel specified in 3.3.2. Other components of the colanders shall be of the material and finish normally furnished by the food mixing machine manufacturer.

3.10.4 Vegetable slicer. When specified (see 6.2), a vegetable slicer capable of being attached to the hub shall be provided. The attachment shall consist of knives enclosed in a housing and arranged to rotate in conjunction with a slicer plate. The attachment shall be capable of slicing, shredding, and grating vegetables, fruits, and nuts. The housing shall be stainless steel, aluminum alloy, or tinned iron or steel, and shall be provided with a hinged plate and operating handle to force the food to be sliced against the knives.

3.10.4.1 Slicer plate. The knife holder and slicer plate shall be of corrosion-resisting metal. Knives shall be of corrosion-resisting cutlery steel, ground and sharpened. The slicer plate for all sizes of food mixing machines shall be not less than 8-1/4 inches in diameter and knives shall be adjustable by external means from 0 to not less than 1/2 inch with respect to the distance from the plate.

3.10.4.2 Accessories for vegetable slicer. When specified (see 6.2) the vegetable slicer attachment shall be furnished with the following accessories as specified, and shall be designed for installation in place of the slicer knife unit:

a. French-fry or julienne cutter, suitable for cuts not less than 3/16 by 3/16 and not more than 1/2 by 1/2 inch.
b. 3/32 inch shredder plate.
c. 3/16 inch shredder plate.
d. 5/16 inch shredder plate.
e. 1/2 inch shredder plate.
f. Extra-fine grating plate.
g. Tubular or equivalent front for use with carrots, cucumbers, celery, or bananas.

3.10.5 Meat and food chopper. When specified (see 6.2), a meat and food chopper capable of being attached to the hub of the mixing machine shall be provided. The chopper shall be complete with manufacturer's standard pan and plunger. Perforation size in plate shall be as specified (see 6.2).

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3.10.6 Dolly bowl adapter. When specified (see 6.2), a dolly bowl adapter to accept the smaller size mixing bowl shall be provided with the model II, sizes 60, 80 and 140 machines.

3.10.7 Adapter. A bowl adapter if necessary to accommodate the smaller bowl as specified in 1.2, shall be provided.

3.11 Navy shipboard installation. For Navy shipboard installation (see 6.2), mounting and incline operation shall be as follows:

3.11.1. Mounting. Model I machines shall have provisions for bolting to bench or counter in accordance with manufacturer's standard practice. Model II machines shall be provided with a base with four holes sized to accommodate the following bolt dimensions:

Machine weight	Bolt diameter (min)
100 - 200 lbs.	3/8 inch
201 - 400 lbs.	1/2 inch
401 - 700 lbs	5/8 inch
701 - 1000 lbs	3/4 inch
1001 - 1500 lbs	1 inch

3.11.2 Inclined operation. The mixing machine shall operate satisfactorily when inclined at an angel of 15 degrees each side of the vertical in each of the two vertical planes at right angles to each other, with no spillage of the product when tested as specified in 4.4.3.

3.12 Finish. The aluminum alloy base and column on models I and II machines shall be given a smooth, natural metal color, ground or burnished finish or any type of polished aluminum finish. Unless otherwise specified (see 6.2), the cast iron or steel base and column of models I and II machines shall be electrotin plated or enameled in accordance with the manufacturer's standard practice and color. All other exposed metal surfaces of the machines (other than agitator shaft and machined moving parts which require lubrication), shall be corrosion-resisting steel or be protected by a nickel or chromium plating.

3.12.1 Exterior surfaces. When specified (see 6.2), for hospital procurements all exterior surfaces for aluminum components shall be ground or burnished or any other type of polished finish and all exterior surfaces for steel components shall be nickel or chrome plating or have a stainless finish in accordance with NSF.

3.12.2 Bowls, rings, and extensions. Bowls, rings and extensions shall have a finish in accordance with NSF.

3.12.3 Hardware. All bolts, nuts, pins, screws, terminals, springs, and other fastenings and fittings shall be of a corrosion-resistant metal or of a metal treated in a manner to render them resistant to corrosion. Steel springs shall be treated to resist corrosion in a manner not to impair their spring qualities.

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3.12.4 Dollies. Dollies shall be given a smooth, natural metal color, finish.

3.13 Workmanship. All components and assemblies of the mixing machine shall be free from dirt and other harmful extraneous material, burrs, slivers, dents, cracks, and rough die, tool and grind marks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges and corners.

3.13.1 Metal fabrication. Metal used in the fabrication on the mixing machine shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be of uniform size and shape.

3.13.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts, or incomplete fusion. All scale be removed from the finish weld area.

3.13.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked, or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.2.2 End item visual examination. The end items shall be examined for defects listed in table II. The lot size shall be expressed in units of end items of one model, size, and style only. The sample unit shall be one end item. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	
Finish	Not finished where required	X	
	Type of finish not as specified	X	
	Blistered, peeled, chipped, scratched, area of no film, runs, sags, or drips (painted surfaces and metallic plating, as applicable.)		X
	Stained or foreign matter imbedded in finish		X
	Any sharp edge, corner, sliver, or burr	X	
	Aluminum castings not polished to a smooth sanitary finish	X	
Design	Any wearing part not accessible for adjustment and replacement	X	
Construction and workmanship (applicable to all components and assemblies)	Any component missing or not as specified	X	
	Any visible component fractured, split, or malformed	X	
	Adjustable assembly that cannot be adjusted	X	
	Adjustable assembly not properly adjusted to perform the function intended	X	
	Casting miscast, cracked, porous, or contains blowholes	X	
	Assembly or component improperly located (for example, gasket improperly fitted to provide seal)	X	
	Plastic guard or housing	X	
Rivets	Rivet(s) missing or broken	X	
	Inadequately peened or loose	X	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Threaded fasteners	Threaded fastener(s) missing, broken, or stripped	X	
Speed selector (gear driven)	Setting not positive at each position of gear selection, including neutral, if applicable	X	
	Marking missing, illegible, not conspicuously located, not permanently affixed, affected by disassembly procedure	X	
Lugs (bowl support and bowl)	Bowl support not provided with three lug positions	X	
	Bowl not provided with lugs to match the lug positions on the bowl support	X	
Stand (when specified)	Not open base welded construction	X	
	Top platform not reinforced	X	
	Edges of top and shelf not flanged or rolled down over frame and secured	X	
	Each leg not provided with an adjustable leveling device	X	
Welding	Missing, incomplete, through burn hole, cracked, severe undercut, or not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Accessories and attachments (as applicable)	Missing or not as specified	X	
Identification marking	Missing, incomplete, or not legible	X	
Lubrication chart and user's instructions	Missing, incomplete, or not legible	X	
	Not affixed as specified	X	

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4.2.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified (see 3.6.9, 3.6.15.2, 3.10.4.1, 3.10.4.2., and when applicable 3.11.1). Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred, shall be 4.0.

4.2.4 End item testing.

4.2.4.1 Representative item testing. One representative end item from the initial lot shall be tested as specified in 4.4.1 through 4.4.9. Any test failure shall be cause for rejection of the lot. Acceptance of any subsequent lot shall be based on satisfactory completion of all tests conducted on the initial lot.

4.2.4.2 Operational testing. Every end item shall be tested as specified in 4.4.1. Any end item exhibiting one or more test failures shall be rejected.

4.2.5 Packaging inspection. An inspection shall be made to determine that the preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-4 and AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. Packaging defects

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of case liners or shrouds; blocking and cushioning inadequate, improper, or missing; bolts improperly affixed.

4.3 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.2, shall be examined and validated as proof of compliance (see 4.3.1 and 4.3.2).

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4.3.1 Underwriters' Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriters' Laboratories standards specified in 3.2 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the mixing machine has been tested and conforms to the applicable requirements of the specified UL standards.

4.3.2 National Sanitation Foundation (NSF). Acceptable evidence of meeting the applicable requirements of the National Sanitation Foundation standards specified in 3.2 shall be one of the following:

a. A listing in the current edition of the National Sanitation Foundation "Listing of Food Service Equipment", and display of the NSF seal on the finished unit.

b. A certification for the unit issued by NSF under their special one-time contract evaluation/certification service.

c. A certified test report acceptable to the contracting officer with the advice of the Army Surgeon General from an independent testing laboratory indicating that the unit has been tested and conforms to the specified NSF standards.

4.4 Methods of inspection.

4.4.1 Operational test. The food mixing machine (and vegetable slicer and meat chopper, if provided) shall be operated without load at high speed for a period of 10 minutes to determine conformance with 3.7.1.2. Spindle speeds shall be checked to determine conformance with applicable requirements of 3.7.1, and 3.7.2. In addition, the gear driven machines shall be operated and the speed selector placed in each speed setting consecutively as well as any neutral position(s) provided, to determine compliance with speed selector requirements of 3.6.2. The speed selector lever shall be placed in the neutral position if one is provided and a force of at least three times the force required to move the lever from neutral position to a speed position, if the interlock were not engaged, shall be applied to the lever in a direction to move the lever into one of the speed positions. The speed selector lever shall not move to a position causing rotation of the drive spindle. This test shall be repeated 12 times for each speed lever setting accessible from the neutral position. The interlock mechanism shall be operated to determine compliance with the requirements of 3.6.2.1. On variable drive machines, the speed selector shall be set at a predetermined speed. The position of the selector shall be marked. At the completion of the test the selector shall not have moved in any direction more than 1/8 of an inch to determine compliance with 3.6.2. Inability of the food mixing machine to pass any of the above specified test, as applicable, shall constitute failure of the machine.

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4.4.2 Dough mixing test. The dough mixing test shall be conducted on mixers in an ambient temperature of 70°F + 5°F.

4.4.2.1 Test preparations. The bowl shall be filled with the proper amount of ingredients necessary to make the maximum rated batch of white bread dough, according to machine size (see 6.2 and applicable subparagraph of 3.7.1 or 3.7.2). The flour shall be hard wheat flour, enriched and bleached, with 11 to 12 percent protein content. The percentages of ingredients in the batch of dough shall be as follows:

Ingredient	Percentage
Flour	57.0
Water	31.3
Yeast (compressed)	1.1
Salt	1.4
Sugar	2.8
Shortening	2.8
Dry skim milk	3.6
Total percentage	100.0

4.4.2.2 Test procedure. A dough hook, of the type and size recommended by the manufacturer, shall be inserted in operating position in the drive spindle of the machine to be tested. The mixing speed on variable drive or three speed mixers shall be the lowest speed. The mixing speed on four speed machines shall be next to the lowest speed. Two full batches of dough shall be mixed and temperature rise of the motor coil windings shall be checked to determine conformance with 3.4. The time interval between finish of the first batch and start of the second batch shall not exceed 30 minutes. Temperature rise shall be the difference between motor temperature just prior to start of the first batch and motor temperature prevailing at finish of second batch. Temperature rise shall be determined by the thermocouple method. The dough mix shall be judged by the appearance, quality, and if possible, by the finished baked product to determine conformance with 3.7.1.4, 3.7.2.2, 3.7.2.3, 3.7.2.4, and 3.7.2.5. During this test, gear driven machines having a neutral speed lever position shall be shifted 12 consecutive times from a speed position to neutral to determine conformance with 3.6.2 and 3.6.2.1. On variable drive machines, the speed selector shall be set at the recommended speed. The position of the selector shall be marked. At the completion of the test, the selector shall not have moved in any direction more than 1/8 of an inch to determine compliance with 3.6.2. After the test, the machine shall be examined for damaged components to determine compliance with 3.4 and 3.6.1. Inability of the food mixing machine to pass any of the above specified tests, as applicable, shall constitute failure of the machine.

4.4.3 Inclined operational test. After the test in 4.4.2.2 is completed position the mixing machine with the base set at an angle of 15 degrees, then operate the mixing machine for 5 minutes each at each side of the vertical in

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each side of the vertical planes at right angles to each other. At each of these positions observe for conformance with 3.11.2. Any nonconformance shall constitute failure of this test.

4.4.4 Mixing bowl support test. The mixing bowl shall be filled with ingredients weighing twice the weight of a full batch of dough, loaded on the bowl support and the adjusting mechanism shall be operated from extreme low position to extreme high position 12 times to determine compliance with applicable requirements of 3.6.11. During this test, for type II machines, the bowl shall be lifted from the dolly. Inability of the food mixing machine to meet the above requirements, as applicable, shall constitute failure of the machine.

4.4.5 Electromagnetic compatibility tests. When electromagnetic compatibility is required (see 3.6.5), the mixing machine shall be tested by the contractor in accordance with test method CE04, CE05 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and the requirements meet MIL-STD-461. Inability of the food mixing machine to pass the electromagnetic compatibility test shall constitute failure of the machine.

4.4.6 Noise level test. The food mixing machine noise level shall be measured in accordance with the field method of ANSI S1.13 except that the A-weighted network shall be used in lieu of the octave or narrow band analyzers to determine conformance with 3.6.8. Any failure to be within the noise limit shall constitute failure of this test.

4.4.7 Bowl capacity test. The bowl shall be filled with water and then emptied into a graduated container to determine compliance with 3.6.9. Inability of the bowl to hold the amount of water specified shall constitute failure of the bowl.

4.4.8 Dolly load test. A bowl of the size the dolly is intended for shall be placed on the dolly and the bowl shall be loaded with weights totalling double the weight of a full batch of dough for a period of 10 minutes. The dolly shall be examined to determine conformance with 3.10.2. Inability of the dolly to operate with specified load shall constitute failure of the dolly.

4.4.9 Table load test (when required for model I, size 20). The table top shall be loaded with an equally distributed load of 250 pounds for 10 minutes. At the end of this period the load shall be removed and the table examined to determine conformance with deformation requirements of 3.7.1.5. Inability of the table to withstand the specified load shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Methods of preservation. All preservation shall be in accordance with the material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.1.1 Cleaning. All unpainted, uncoated, exposed ferrous metal surfaces, except corrosion-resisting steel surfaces of the food mixing machines, detached parts, accessories, and attachments shall be cleaned (process C-1) and thoroughly dried.

5.1.1.1.2 Preservative application. All cleaned surfaces coming in contact with food shall be coated with type P-14 preservative. Ferrous metal shafts and pulleys shall be coated with primer, conforming to TT-P-664, the primer dried, and belts replaced. All other cleaned surfaces shall be coated with type P-6 preservative. All bearings and gears shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.1.3 Protection of preservative. Surfaces, from which preservatives might be removed by contact with packing materials, shall be covered, wrapped, or bagged with greaseproof barrier material conforming to type I or II, grade A, class 1 or 2 of MIL-B-121. Barrier material shall be secured with twine made of jute, sisal, manila, or cotton, or with tape conforming to type III, grade A of PPP-T-45, or heat sealed when heat sealable barrier material is used.

5.1.1.1.4 Sealing. Exterior switches, terminals, control boxes, and all openings in exterior mounted motors shall be sealed with tape conforming to PPP-T-60.

5.1.1.2 Unit packing. Accessories, attachments, detached parts, mixing bowl, agitators, wire whip, and dolly, prepared as specified, shall be unit packed in a snug fitting fiberboard, nailed wood or cleated plywood box or boxes conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636, except that dimensional and weight limitations shall not be waived; class 1, style 2 or 4, grade B of PPP-B-621, or domestic type, style A or B, grade B of PPP-B-601, respectively. The contents shall be unit packed, cushioned, blocked, or braced within the box in accordance with the manufacturer's standard practice to prevent movement while in transit. Each box shall be securely closed in accordance with the appendix of the box specification. Only when the unit pack cube can be reduced shall the mixing bowl be secured in place on the mixing machine.

5.1.2 Commercial. Each mixing machine shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

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5.2.1.1 Machines together with accessories weighing more than 500 pounds. Each machine together with accessories, preserved as specified in 5.1, shall be packed in a crate conforming to type I, class 1 or 2, style a of MIL-C-104. The box(s) containing accessories, and detached parts, shall be placed in an unused portion of the crate. The machines shall be secured with minimum 3/8 inch diameter carriage bolts by bolting the machine at all four corners to the base of the crate and the accessory boxes secured by lumber blocking and bracing, in accordance with the appendix of the crate specification.

5.2.1.2 Machines together with accessories weighing more than 65 pounds. Each machine together with accessories weighing more than 65 pounds and not exceeding 500 pounds, preserved as specified in 5.1, shall be packed in a nailed wood shipping container conforming to class 2, style 2 or 4, grade A or B, type 3 load of PPP-B-621, except that minimum thickness of the bottom members of the shipping container shall be 3/4 inch. Models I and II machines shall be secured by bolting to the base of the container. Each shipping container shall be provided with skids, in accordance with the container specification, except that skids shall be positioned, or additional skids used, as required, to insure that all supporting legs of the machine are positioned over a skid and to engage the bolts used to secure the machine to the base of the shipping container with bolts passing through the skids. When the base of the machine has threaded mounting holes, machine bolts shall be used and the bolts shall be recessed so as not to project below the skids. When model II machines are secured by less than four bolts, the machine shall be further secured with lumber blocking and bracing, as specified herein, with cushioning used between the lumber blocking and the machine. The contents shall be waterproofed as specified in 5.2.1.2.1. The box or boxes containing accessories and detached parts, shall be secured in an unused portion of the shipping container with lumber blocking and bracing. The blocking and bracing shall not depend on end grain nailing alone, but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. Each shipping container shall be closed and strapped in accordance with the appendix of the container specification.

5.2.1.2.1 Waterproofing. Each machine shall be covered with a full height shroud made of waterproof barrier material conforming to class E-2, H-5 or M-1 of PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260. Fiberboard boxes containing accessories shall be wrapped with waterproof barrier material and the wrap sealed as specified herein for shrouds. Nailed wood boxes containing accessories shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547.

5.2.1.3 Machines together with accessories weighing 65 pounds and less. Each machine together with accessories, preserved as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The contents shall be immobilized by means of fiberboard inserts. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

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5.2.2 Level B packing.

5.2.2.1 Machines together with accessories weighing more than 500 pounds. Each machine together with accessories, preserved as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the waterproof liner shall not be required, and the machine shall be covered with a shroud fabricated from minimum 0.002 inch thick polyethylene.

5.2.2.2 Machines together with accessories weighing more than 65 pounds. Each machine together with accessories weighing more than 65 pounds and not exceeding 500 pounds, preserved as specified in 5.1, shall be packed as specified in 5.2.1.2, except that the shipping container shall conform to class 1, style 2 or 4, grade B, type 3 load of PPP-B-621 and waterproofing of contents shall not be required. The machine shall be covered with a shroud fabricated from minimum 0.002 inch thick polyethylene.

5.2.2.3 Machines together with accessories weighing 65 pounds and less. Each machine, together with accessories preserved as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.3.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Commercial packing. Mixing machines preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Civil Agencies. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with FED-STD-123 or ASTM D 3951 as applicable.

5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The food mixing machines are intended for use in bakeries and kitchens. Machines are intended for mixing bread and roll doughs, and can be used for lighter mixing and whipping. Machines may be equipped with optional accessories for preparing meats and vegetables. The storage rack referenced in 6.3 is intended for storing mixing bowls and other accessories.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Model, size, and style required (see 1.2).
c. Electrical characteristics, if other than specified (see 3.6.4.1).
d. Motor characteristics, if other than specified (see 3.6.4.1).
e. Cord and attachment plug, if other than specified (see 3.6.4.3).
f. When electromagnetic compatibility is required (see 3.6.5).
g. When fungus and moisture treatment of electrical components is required (see 3.6.6).
h. Capacity and quantity of additional bowls, when required (see 3.6.9).
i. When a plastic cover is required for 80 quart or 140 quart size mixing bowl (see 3.6.9).
j. Spindle speeds, if other than specified (see applicable subparagraphs of 3.7.1 and 3.7.2).
k. Method of bolting model I machines, if other than specified (see 3.7.1.2).
l. When table is required for model I, size 20 mixing machine (see 3.7.1.5).
m. When table with casters is required (see 3.7.1.5).
n. When hub drive is not required (see 3.9).
o. When a dolly is not required for model II, sizes 60, 80, and 140 mixing machines (see 3.10.2).
p. When an additional bowl is required for the colander (see 3.10.3).
q. When a vegetable slicer and slicer accessories are required (see 3.10.4, 3.10.4.2).
r. When a meat and food chopper is required (one unit required for kitchen use (see 3.10.5).
s. Size of perforations for food chopper plate when provided (see 3.10.5).
t. When a dolly bowl adapter is required (see 3.10.6).
u. When mixing machine is intended for Naval shipboard installation (see 3.11).
v. Finish of cast iron or steel base and column of models I and II machines, if other than specified (see 3.12).
w. When procurement is for hospitals (see 3.12.1).
x. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
y. When weather-resistant fiberboard is required for level B pack (see 5.2.2.3.1).

6.3 Rack storage. The mixing bowl storage rack used for component storage (see 6.1) is covered by MIL-R-43870.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

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OO-M-0038K(Army-GL)

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User activity:

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Preparing activity:

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Project No. 7320-A779

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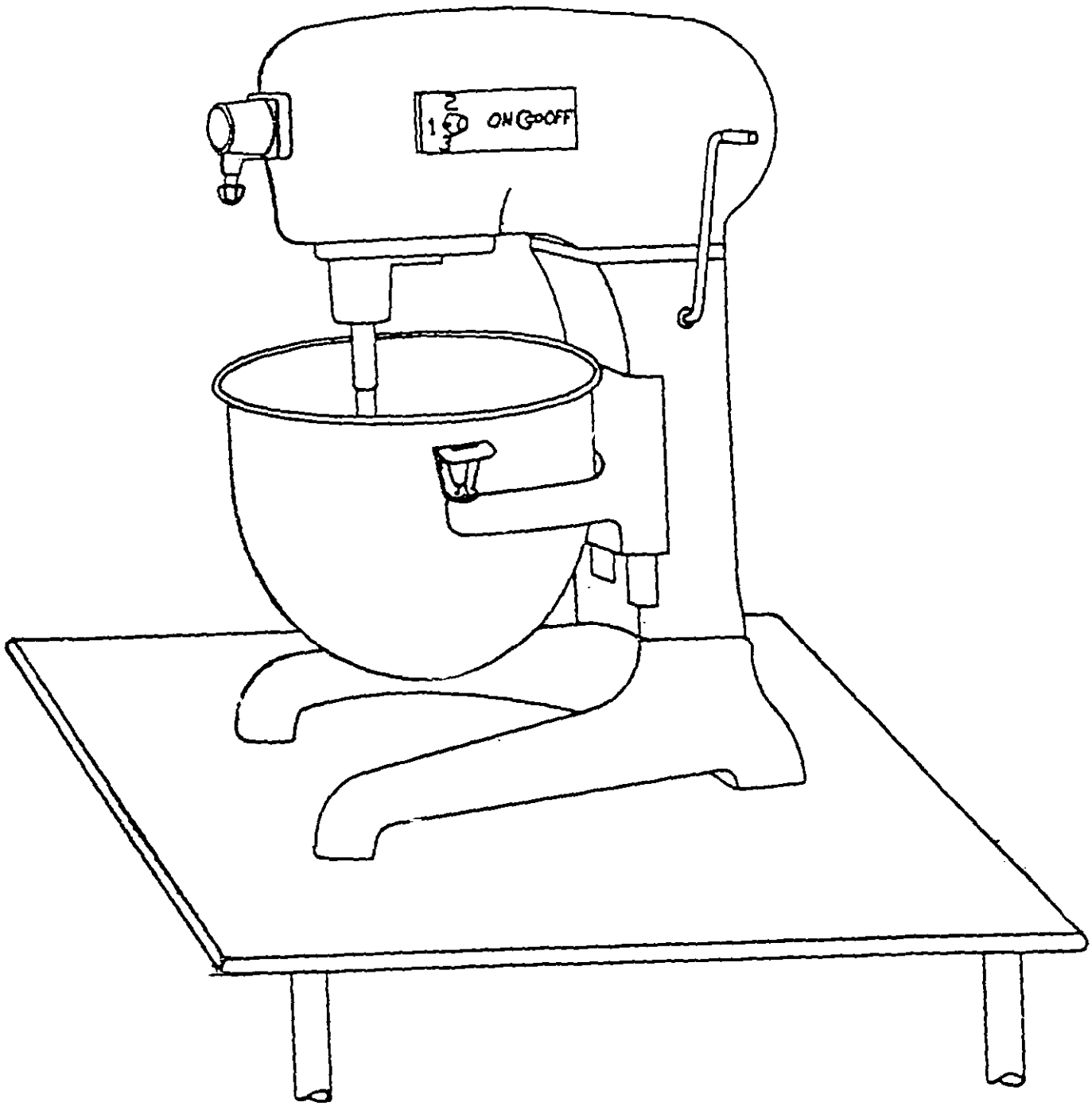


FIGURE 1. MODEL I, BENCH TYPE FOOD MIXING MACHINE

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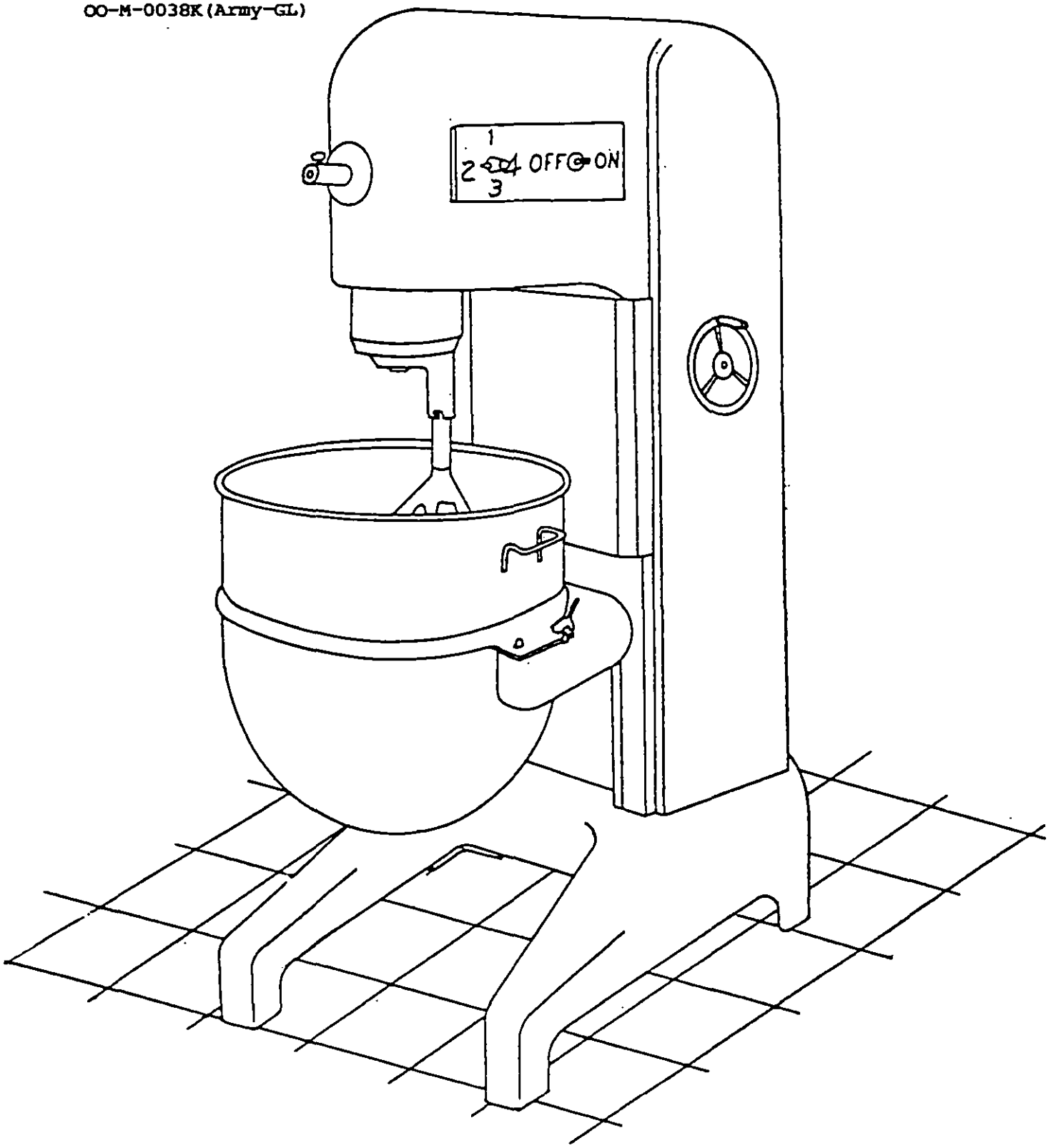


FIGURE 2. MODEL II, FLOOR TYPE FOOD MIXING MACHINE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
OO-M-0038K(Army-GL)	Mixing Machines, Food, Electric (Vertical, Commercial Type, Bakery and Kitchen)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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